

Halloran Dairy

Both papers recommended the same plant. They disagreed only about how much of it to build, and that turned out to be the whole question.

Halloran Dairy processed fluid milk and cultured products in three plants across Wisconsin and Minnesota. The proposal in front of the capital committee was a high-protein strained yogurt line — a category growing at better than twenty percent a year, in which Halloran had no position and two of its three largest customers had asked it to have one.

The uncertainty was not whether the category would grow. It was whether Halloran would get shelf space. Two national brands already held most of it, and the private-label programs that Halloran would supply were being bid in a cycle that would not conclude until the middle of 2028. Ines Kovalenko, who ran category strategy, put the odds that Halloran wins meaningful placement at "somewhere around forty-five percent, and anyone who tells you they know it better than that is selling something."

The engineering group had costed two ways of building it.

Build it now. Three filling and straining lines installed at once in the Eau Claire plant, \$36 million, ready for the 2027 season. Shared construction, one mobilization, one commissioning. Years 1 and 2 would run at \$6.0 million of EBITDA while the category developed and the bid concluded, because the capacity would be there whether the volume was or not.

Stage it. One line now, \$18 million. Years 1 and 2 at \$3.6 million of EBITDA. Then, at the end of 2028 with the bid decided, either add the second and third lines for \$22 million, or leave the single line as it stands and run it to the end of the horizon. Building the two later lines separately cost \$4 million more than building all three at once — a second mobilization, a second commissioning, and a plant that has to keep running while the work goes on.

What happens in 2029 depends on the award, and so does what a single line is worth. While the bid is open, the private-label volume is unallocated and everyone in the category picks some of it up: Halloran had been quoted bridge business by both of the customers that asked it to enter, which is why one line covers \$3.6 million of EBITDA in 2027 and 2028 without any shelf space at all. That volume goes to the winners when the award lands. A line that wins runs full at the program margin; a line that loses is left with residual co-manufacturing and whatever spot business it can find.

So from 2029 onward, with the shelf space, one line generates \$5.0 million of EBITDA a year and three lines generate \$15.0 million — the lines are full either way, and three of them simply fill three times over. Without the shelf space, one line generates \$2.0 million and three lines generate \$4.0 million, because the volume to fill the second and third lines does not exist at any price Halloran would accept.

Nobody was going to buy the line off Halloran. The committee had asked about it and the answer was that a single installed strained-yogurt line in Eau Claire is worth what it produces and nothing more; there is no second-hand market for a plant in a category where the only buyers are the two brands that already won. So the single line, once built, would be run for the eight years whether the bid landed or not.

"So the question is whether we pay four million dollars for the right to change our minds," Kovalenko said. "Which is a strange thing to put in a capital request, because there is no line on the form for it."

EXHIBIT 1 — THE TWO PATHS

	BUILD IT NOW	STAGE IT
Investment, year 0	\$36.0m	\$18.0m
Investment, end of year 2, if expanded	—	\$22.0m
EBITDA, years 1–2	\$6.0m	\$3.6m
EBITDA years 3–8, shelf space won	\$15.0m	\$15.0m expanded, \$5.0m one line
EBITDA years 3–8, shelf space lost	\$4.0m	\$4.0m expanded, \$2.0m one line
Salvage, end of year 8, shelf space won	\$14.0m	\$14.0m
Salvage, end of year 8, shelf space lost	\$5.0m	\$4.0m

Probability of winning meaningful shelf space, resolved at the end of year 2: 45 percent.

All equipment is seven-year MACRS property. Halloran's tax rate is 25 percent and its cost of capital is 11 percent. The project horizon is eight years.

EXHIBIT 2 — SEVEN-YEAR MACRS

YEAR OF SCHEDULE	1	2	3	4	5	6	7	8
Percentage	14.29	24.49	17.49	12.49	8.93	8.92	8.93	4.46

The schedule runs one year longer than the class life because of the half-year convention. It ignores salvage value: the asset is written down to zero regardless of what it is expected to fetch.

ASSIGNMENT

- 1 Value Build It Now. Compute its NPV in the shelf-space-won state and in the shelf-space-lost state separately, then take the expectation.
- 2 Value Stage It, working backward. Start at the end of year 2 and value both of the things Halloran could do there — expand, or run the single line to the end of the horizon. Only then discount back.
- 3 Which path is worth more, and by how much? State in one sentence what Halloran is buying with the extra \$4 million.
- 4 Stage It is a one-line project plus the right to expand. Value the one-line project on its own — one line, held to year 8, no expansion ever — and take the difference. How much of Stage It's value is the line, and how much is the right? Be careful about what the option value you compute is already net of.
- 5 Kovalenko's 45 percent is a judgment, not a measurement. Find the probability at which the committee should be indifferent between the two paths, and say which side of it you think Halloran is on.

Before class

- READ** Eight minutes is enough. Come with a number, not a summary.
- BRING** Your cash flow schedule, your NPV, and the one assumption you are least sure of.
- EXPECT** Twenty minutes of argument. More than one answer is defensible; an undefended one is not.